Umer Niazi

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Education

University of Engineering and Technology, Lahore — Lahore, Pakistan
BS in Artificial Intelligence — *2025 – 2029 (Expected)*

Forman Christian College — Lahore, Pakistan
Intermediate in Computer Science (ICS) — *2022 – 2024*

Projects

Thermal Capital [code] | *Python, FastAPI, React, TypeScript, Geospatial Processing, SQLite* — Aug 18 – 30, 2026

- Developed a decision-support prototype during a 12-day hackathon for municipal heat-intervention planning, pairing geospatial processing with constrained optimization to maximize urban cooling within capital budget limits
- Engineered a FastAPI REST API and interactive React/TypeScript interface to simulate cooling interventions, evaluate spatial coverage, and optimize multi-asset municipal budgets
- Implemented 112 automated unit and integration tests across data models, knapsack optimization algorithms, and API endpoints to ensure system correctness and reproducibility

News NLP Pipeline [code] [demo] | *Python, BERTopic, spaCy, Transformers, Streamlit, SQLite* — May – July 2026

- Built an end-to-end NLP pipeline for unsupervised topic discovery, sentiment analysis, and named-entity extraction across 15 years of Dawn News headlines without manual labeling
- Scraped and preprocessed 350K+ headlines and 420K+ entity mentions using BERTopic and spaCy; modeled 20 distinct topic clusters and deployed an interactive Streamlit dashboard

Himalayan Wildlife Classifier [code] [demo] | *Python, PyTorch, ResNet18, Streamlit* — May – July 2026

- Built an image classifier for Snow Leopard, Markhor, and Himalayan Brown Bear, plus a background Other class, using transfer learning on a frozen ResNet18 backbone
- Achieved 96.55% validation accuracy; implemented an 80% confidence threshold for selective prediction to filter ambiguous inputs and deployed an interactive Streamlit demo

wmus [code] | *Python, windows-curses, pygame* — Jan – June 2026

- Built a lightweight terminal-based music player for Windows inspired by the cmus workflow, featuring keyboard-first navigation, a custom curses interface, and configurable keybindings
- Implemented fuzzy track search and metadata caching for fast startup across large audio collections; packaged as a standalone installer using PyInstaller and Inno Setup

Technical Skills

Languages: Python, TypeScript, JavaScript, C++, SQL, GDScript
AI & Machine Learning: PyTorch, scikit-learn, Transformers, Computer Vision, NLP
Software & Systems: FastAPI, React, Streamlit, SQLite, Linux, Git, Bash
Game Dev: Godot, Unity